

# AMEY SHUKLA

## PERSONAL INFORMATION

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## GOAL

Advance practical cryptography by developing theoretical tools for authentication and privacy that address challenges in building resilient systems. Enable non-experts to implement resilient systems effectively.

## EDUCATION

|                                            |                                                                                                              |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| <i>PhD in Computer Science</i>             | <i>2023–2028</i> University of Connecticut<br>In the area of applied cryptography under Dr. Benjamin FULLER. |
| <i>Masters in Computer Science, Thesis</i> | <i>2021–2023</i> Wichita State University<br>In the area of cybersecurity under Dr. Sergio Salinas MONROY.   |
| <i>Bachelors in Computer Science</i>       | <i>2017–2021</i> Wichita State University<br>Dual/Accelerated program under Dr. Sergio Salinas MONROY.       |

## WORK EXPERIENCE

|                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>University of Connecticut</i> | <i>2023–Present</i> Graduate Research Assistant<br>Working on exciting problems in applied cryptography, such as noisy authentication, oblivious computation, and searchable encryption.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <i>Wichita State University</i>  | <i>2021–2023</i> Graduate Teaching Assistant<br>TA'd: Introduction to Cybersecurity, Routing & Switching I, Applied Computing & Networks, Applied Programming & Scripting, Applied Web Apps & Database Development, as well as designed & taught a brand new CTF prep course from scratch (now a required course for applied computing cybersecurity track degree).<br><br>Responsibilities: Graded assignments, reached out and guided struggling students, reinforced material, answered student questions during office hours, and tracked attendance                                                                                                                                |
| <i>NetApp</i>                    | <i>2019–2021</i> Software Engineer Intern<br>NetApp, formerly Network Appliances, is a storage appliance company that focuses on SAN, NAS, and other hybrid-cloud storage solutions. My work was mostly written in Java, Python, and C++.<br><br>Developed and maintained E-Series REST API, which served as an interface to low-level details of the storage appliance.<br><br>Developed a next-generation container execution framework utilizing Kubernetes, Docker, and Ansible to run storage software and functions on corresponding software with load balancing and highly available topology.<br><br>Developed a resource provisioning framework to provision storage software |

builds virtually on demand, saving many hours of developer downtime.

Implemented stress tests for storage appliances to test recovery in case of faults.

## PUBLICATIONS

**Note:** Many are alphabetical by last name, as is the convention for cryptography research.

- Eprint* Noah Arce-Caliskan, Benjamin Fuller, Gadalia Montoya Weinberg O'Bryan, Maryam Rezapour, Amey Shukla.  
*Sample-then-Correct: Practical fuzzy extractors for biometrics beyond the iris.* ePrint.
- Eprint* Benjamin Fuller, Ariel Hamlin, Arinjita Paul, Maryam Rezapour, Ronak Sahu, Amey Shukla, Mason Stuart.  
*Updatable Oblivious Key Value Stores with Access Control and Application to Multi Key Searchable Encryption.* ePrint.
- Eprint* Mahdieh Heidarpour, Sage Pia, Maryam Rezapour, Amey Shukla, Gagan Agrawal, Hoda Maleki, Benjamin Fuller.  
*Block Oblivious Single Access Machines and Oblivious Graph-Based Approximate Nearest Neighbor Search with Small Client Storage.* ePrint.
- Eprint* Sage Pia, Ananya Appan, Maryam Rezapour, Amey Shukla, Nikhil Date, Benjamin Fuller, Ling Ren, David Heath.  
*Oblivious Single Access Machines are Concretely Efficient.* ePrint.
- ACM CCS Amey Shukla, Luke Demarest, Benjamin Fuller, Sohaib Ahmad, Caleb Manicke, Alexander Russell, Sixia Chen.  
*Fuzzy Extractors are Practical: Cryptographic Strength Key Derivation from the Iris.* ACM CCS 2025.
- Eprint* Benjamin Fuller, Arinjita Paul, Maryam Rezapour, Ronak Sahu, Amey Shukla.  
*MARS: Low-Leakage Multi Adversarial Owner and Reader Replication-free Searchable Encryption from Private Information Retrieval.* ePrint.

## OTHER INFORMATION

- Awards* 2026 · UConn Computer Science Predoctoral Fellowship.
- Awards* 2025 · ACM CCS Travel Award.
- Awards* 2025 · UConn Computer Science Predoctoral Fellowship.
- 2025 · UConn Fall Conference Participation Award.
- 2023 · UConn CSE Fellowship.
- 2021 · Global Select Scholarship.
- Invited Talks* Sample then Correct.
- New England Security Day 2026 · Talk about amplification for fuzzy extractors at the University of Massachusetts Amherst.
- Fuzzy Extractors are Practical.
- Google · Talk about Fuzzy Extractors at Google.
- Microsoft Research · Talk about Fuzzy Extractors at Microsoft Research.
- JP Morgan · Talk about Fuzzy Extractors at JP Morgan.
- Boston University · Talk about Fuzzy Extractors at BU Crypto Seminar.
- Persona · Talk about Fuzzy Extractors at Persona.

## OTHER EXPERIENCES

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|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Volunteer</i>                               | <div> <div>2017–Present</div> <div>Volunteer</div> </div> <p>Invited to help moderate a cybersecurity summer camp for children in High School. Helped children during workshop, answered questions, debugged their code, and tasks to point out issues.</p> <p>Volunteered at a program that helped university freshman make the best of their first semester.</p> |
| <i>Wichita State<br/>Linux Users<br/>Group</i> | <div> <div>2019–2021</div> <div>Vice President</div> </div> <p>Managed a Linux Users Group with 100+ members, and led workshops on applied cryptography, security, systems programming, ansible, computer graphics, and of course Linux.</p>                                                                                                                       |
| <i>Keywords</i>                                | <p>C · Python · Git · L<sup>A</sup>T<sub>E</sub>X · Docker · Ansible · Kubernetes · LXC · KVM · Mercurial · Cryptography.</p> <p>Can write most imperative programming languages with access to documentation.</p>                                                                                                                                                 |

September 13, 2026